

Cheatsheet

Global conventions

- Random variables are capital letters (e.g., X, S, R); values are lowercase (e.g., x, s, r).
- Finite/discrete setting throughout unless explicitly stated.
- $\log \equiv \log_2$. Units: bits. Natural logs use $\ln$ and give nats.
- A probability mass function (PMF) $p_X(x)$ satisfies $p_X(x) \geq 0$ and $\sum_x p_X(x) = 1$.

Sample spaces, PMFs, and expectations

- Finite set (sample space) Ω ; cardinality $|\Omega|$.
- Random variable $X : \Omega \rightarrow \mathcal{X}$ and its PMF $p(x) = \Pr(X = x)$, where $\int_x p(x) = 1$ always holds and $\Pr(X \in A) = \int_{x \in A} p(x)$
- Expectation: $\mathbb{E}[f(X)] = \int_{x \in \mathcal{X}} f(x) p(x)$.
- Variance: $\text{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2$.

Joint, marginal, conditional; Bayes; independence

Objects:

- Joint PMF $p(x, y) = \Pr(X = x, Y = y)$.
- Marginals: $p(x) = \sum_y p(x, y)$, $p(y) = \sum_x p(x, y)$.
- Conditional: $p(x | y) = \frac{p(x, y)}{p(y)}$ (when $p(y) > 0$).
- Bayes theorem:

$$p(x, y) = p(x | y)p(y) = p(y | x)p(x) \quad (1)$$

$$p(x | y) = \frac{p(y | x)p(x)}{p(y)} \quad (2)$$

- Independence: $X \perp Y \iff p(x, y) = p(x)p(y) \quad \forall x, y \iff p(x | y) = p(x) \text{ for all } y \text{ with } p(y) > 0$.

Logarithms and “bits” intuition

- Log rules: $\log(ab) = \log a + \log b$, $\log(a/b) = \log a - \log b$, $\log(a^k) = k \log a$.
- Change of base: $\log_b x = \frac{\log x}{\log b}$.
- Bits intuition: $\log_2 N$ is the number of yes/no questions needed to identify one of N equiprobable outcomes.